Zachary Tianyi Tang

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Research Interests

Sample-efficient robot learning, human-in-the-loop adaptation, and precise/deformable-object manipulation.

Technical Skills

Programming: Python, C, JavaScript, TypeScript
Robotics / ML Libraries: PyTorch, JAX, OpenPI, LeRobot, Isaac Sim, Isaac Lab, OpenCV
Systems / Tools: CUDA, TensorRT, SLURM, Docker, Git, Linux, Bash
Web / Cloud: React, FastAPI, Firebase, Supabase, GCP (Cloud Run, BigQuery), AWS (SageMaker, Bedrock)

Research & Engineering Experience

Dynamic Optimization & RL Lab, University of Toronto (Prof. Chi-Guhn Lee) Toronto, ON
Undergraduate Researcher *Sep 2025 - Present*

- Co-authored **WireCraft (CoRL 2026 Spotlight; DOM-R3 @ IROS 2026 Oral)**, a **flexible-wire manipulation benchmark** showing that simulated plus real demonstrations outperformed real-only training on physical robots.
- Deployed, fine-tuned, and evaluated $\pi_{0.5}$ **vision-language-action policies** with **OpenPI**, plus **ACT and multi-task DiT policies**, for UR5 wire manipulation; built **LeRobot-compatible datasets** and scalable training and inference pipelines.
- Built and benchmarked **Isaac Lab environments** for articulated and deformable wires, using **reinforcement learning with PPO**, reward shaping, clip-routing demonstrations, and task-success metrics for robot manipulation.
- Investigated rare **PhysX contact failures** across millions of RL steps with joint-state logging and **Three.js replay**; traced angular-velocity spikes to unstable contacts, informing mesh changes that made training substantially more stable.

Intelligent Robot Learning Lab, University of Alberta (Prof. Matthew Taylor) Remote
Undergraduate Researcher *Sep 2025 - Present*

- Led an undergraduate research team studying how AI agents coordinate when each agent has incomplete information, building **JAX/JaxMARL training environments** and communication experiments.
- Built an **Isaac Lab simulation** with two **Franka robot arms** to study how robots share limited information while coordinating tasks, integrating high-level action controllers, local observations, and learned messaging.

RLLAB, Yonsei University (Prof. Youngwoon Lee) Seoul, South Korea
Research Intern *May 2026 - Aug 2026*

- Led research on helping robots learn from less **human teleoperation data**, studying how to reduce time spent collecting it and make better use of each demonstration and intervention.
- Built and operated a **dual-arm 14-DoF YAM teleoperation setup** and helped develop its control framework for **DAgger-style intervention collection**, gathering real-robot operator demonstrations and human interventions.
- Integrated **real-time chunking for diffusion policies** to blend overlapping action chunks for asynchronous robot control while maintaining policy performance.
- Developed and evaluated **offline post-training (RL-Token-style)** using failure and intervention data, earning **top-8 finishes** in both tracks of the **RSS 2026 Robotics Foundation Models Challenge**.

Education

University of Toronto Toronto, ON
Honours B.Sc., Computer Science; GPA: 3.77/4.00 *Sep 2024 - Jun 2028*

- **Award:** People's Choice Award, 2026 Canadian Undergraduate Conference on AI (CUCAI)

Projects & Leadership

Heimer Academy (Devpost) (GitHub)

- **1st Place — AWS × Riot Games Rift Rewind ($10,000 USD + $10,000 AWS credits)**
- Built a **League of Legends champion recommendation system** with **Amazon SageMaker and Amazon Bedrock**, using gameplay patterns across **10,000+ player profiles** and similarity analysis over **600+ champion abilities**.

University of Toronto Machine Intelligence Student Team (UTMIST) Toronto, ON
Director of Academics / Director of Research Collaborations *Jun 2025 - Sep 2026*

- Established a research collaboration with the **University of Alberta** and led **15+ researchers** across two AI research teams.
- Led **weekly video-upscaling meetings**, coordinating model evaluation, GPU optimization, and OBS integration.